

1



nuScenes source data

- 3,741 CAM_FRONT keyframes
- 94 verified scenes

2



Generator-matched variants

- 1 null edit + 12 trigger edits
- 13 variants per source
- 48,633 total variants

3



Scene-grouped split first

- 65 train scenes: 33,618
- 14 validation scenes: 7,228
- 15 test scenes: 7,787

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Leakage-safe evaluation

- controlled main
- strict category LOTO
- fixed-budget diversity

Common evaluation protocol



Generator-matched control

Same editor, resolution, inference settings, and generation procedure for both labels.

Partition roles



Training: detector-head fitting



Validation: checkpoint + threshold



Test: untouched final evaluation



Integrity constraints

- All derivatives of a scene stay in one split.
- Zero scene and source-frame overlap.
- Strict LOTO excludes the held-out trigger category from training and validation.



Fixed-Budget Diversity Audit

For k in $\{2, 4, 6, 8, 11\}$, trigger-category diversity is varied while keeping 5,000 training positives, 1,000 validation positives, and fixed negative pools.